TopoCircuit-SAE: Frequency-Ordered DAG Routing to Measure and Remove Multi-Parent Feature Absorption in Sparse Autoencoders

Farhan Aaqil Durrani
Final-Year B.Tech AI/ML, JPNCE Mahbubnagar · AI Engineer Intern @ Sydney Leads Digital Marketi
`topocircuit-sae-v1.0-prereg-phase5`

November 2026

Abstract

Sparse Autoencoders (SAEs) suffer from *feature absorption*, a structural pathology in which a narrow, specific latent (e.g., `"dog"`) absorbs the decoder direction of a broader parent latent (e.g., `"animal"`), causing the parent latent to fail to fire on tokens belonging to its child concept. While hierarchical SAEs such as Matryoshka SAE, H-SAE, HSAE, and Tree SAE mitigate absorption on single-parent trees, real-world concepts routinely inherit from multiple simultaneous parents (e.g., `"bat"` $\rightarrow$ {`mammal`, `flier`}; `"Paris"` $\rightarrow$ {`Letter P`, `noun.location`, `TitleCase`}). Moreover, existing evaluation suites such as SAEBench score absorption solely on the single-parent first-letter task, and the only prior multi-parent method (AG-SAE) induces edges via a discrete post-hoc combinatorial search loop. In this paper, we introduce **TopoCircuit-SAE**, comprising: (1) **MPAbs**, a 112-cell multi-parent feature absorption benchmark and mixed-topology synthetic generator with exact ground truth; (2) **Lemma 1** and **Proposition 2**, proving that ranking latents by firing frequency guarantees an acyclic multi-parent graph without continuous DAG penalties and that flat SAEs selectively absorb parents in strictly increasing order of conditional magnitude variance $\sigma_i^2 < \lambda$; and (3) **DR-SAE**, an end-to-end frequency-ordered DAG-routed SAE whose nilpotent ancestral router $z_{\text{eff}} = (I - A)^{-1}z = \sum_{h=0}^{H} A^h z$ activates all ancestors at zero native L_0 cost. Across 3 seeds on Pythia-160M, DR-SAE reduces mean MPAbs by **88.3%** relative to Matryoshka SAE at matched native $L_0 = 50.0$ (3.99% vs. 34.05%) and by **80.3%** at strictly matched effective $L_0 = 50.0$ (6.71%, paired Wilcoxon $p = 8.19 \times 10^{-19}$), while increasing cross-seed edge reproducibility from 36.4% (AG-SAE-lite) to **78.4%** and shrinking 0.80-faithful Sparse Feature Circuits on Pythia-70M subject-verb agreement by **39.0%**.

1 Introduction and Honest Positioning Against Prior Work

Sparse autoencoders decompose dense language model activations $x \in \mathbb{R}^d$ into sparse, overcomplete linear combinations of interpretable feature directions [14, 15]. However, Chanin et al. [1] demonstrated that whenever ground-truth concepts form a hierarchy $p \rightarrow c$, the sparsity penalty incentivizes the child latent c to fold the parent direction u_p into its own decoder atom and suppress z_p on child tokens.

Recent work has proposed structured dictionary architectures—including Matryoshka SAE [2], H-SAE [7], HSAE [6], Tree SAE [5], HiPACE [4], and AG-SAE [3]. Despite rapid progress in 2025–2026, three fundamental gaps remain open:

1. **The Measurement Gap (Single-Parent Bias):** Virtually every published absorption benchmark (including SAEBench [9]) evaluates absorption exclusively on the *first-letter task*, where every token has exactly one parent. A tree-prior SAE can score near-zero absorption on first letters by routing each word to its orthographic initial, while silently absorbing its semantic and morphological parents.
2. **The Method Gap (Discrete Search vs. End-to-End Acyclic Routing):** Tree-structured SAEs restrict each latent to at most one parent ($|\text{Pa}(c)| \leq 1$), even though AG-SAE [3] found that 23.55% of parented features in Gemma-2-2B require multiple parents. Yet AG-SAE constructs its graph via discrete induce-then-retrain outer loops with greedy cycle removal rather than learning edge weights as differentiable parameters guaranteed acyclic by construction.
3. **The Circuit Gap (Unmeasured Fragmentation):** Although absorption is widely hypothesized to fragment causal feature circuits across dozens of narrow child latents, no controlled study has tested whether reducing multi-parent absorption yields smaller faithful circuits on language tasks [11].

Core Contributions of TopoCircuit-SAE: We close all three gaps within a pre-registered, falsifiable experimental protocol: (1) the 112-cell **MPAbs** benchmark and mixed-topology synthetic generator; (2) **DR-SAE**, an end-to-end frequency-ordered DAG-routed SAE; (3) **Proposition 2**, predicting which parent is absorbed first under multi-parent sparsity pressure; and (4) a controlled 6-layer circuit compactness study on Pythia-70M.

2 Theoretical Framework: Acyclicity and Selective Absorption

Let $\mathcal{G}^* = (V^*, E^*)$ be a ground-truth concept DAG where each concept $i \in V^*$ has a unit-norm residual atom $u_i \in \mathbb{R}^d$ satisfying hierarchical orthogonality $\langle u_c, u_p \rangle = 0$ for all $(p \to c) \in E^*$ [8]. Whenever child c fires on token t ($s_c(t) > 0$), all parents $p \in \mathrm{Pa}^*(c)$ fire simultaneously ($s_p(t) > 0$).

Lemma 1 (Frequency-Ordered Acyclicity and Partial Coverage Bound)**.** *Let $f_i = \Pr(z_i > 0)$ denote the firing frequency of feature i, and suppose parent p fires on child c's active tokens with conditional coverage $\tau = \Pr(p \text{ active} \mid c \text{ active}) \in (0, 1]$ and background rate $\beta = \Pr(p \text{ active} \mid c \text{ inactive}) > 0$. Then:*

$$f_p = \tau f_c + (1 - f_c)\beta. \tag{1}$$

Under exact coverage ($\tau = 1$) and $\beta > 0$, $f_p > f_c$ holds strictly. Therefore, constraining candidate directed edges $(p \to c)$ to pairs satisfying $f_p > f_c$ (with ties broken by latent index) guarantees that the adjacency matrix A is strictly upper-triangular in frequency rank, nilpotent ($A^H = 0$ for $H \geq |V|$), and acyclic by construction while preserving 100% of exact-coverage edges. Under partial coverage $\tau < 1$, an edge $(p \to c)$ is dropped only if $\beta \leq \frac{(1-\tau)f_c}{1-f_c}$ (the near-duplicate regime).

Proposition 1 (Sparsity-Neutral Ancestry Removes the Absorption Incentive)**.** *Under ancestral routing $z_{eff} = (I - A)^{-1} z$ and reconstruction $\hat{x} = R z_{eff} + b_{dec}$, firing a single native child code $z_c > 0$ automatically activates all ancestors $p \in \mathrm{Anc}(c)$ via $z_{eff,p} = [(I - A)^{-1}]_{p,c} z_c$ at zero additional native L_0 cost. Combined with the hierarchical orthogonality penalty $\lambda_\perp \sum_{(p,c)\in E} \langle r_p, r_c \rangle^2$, the unique global minimizer recovers pure orthogonal residual atoms $r_c = u_c, r_p = u_p$ with zero parent false negatives.*

Proposition 2 (Selective Multi-Parent Absorption Phase Boundary)**.** *Let child c have $m \geq 2$ pairwise orthogonal parents $\mathrm{Pa}^*(c) = \{p_1, \ldots, p_m\}$ whose ground-truth magnitudes conditioned on c being active ($s_c = 1$) have conditional means $\mu_i = \mathbb{E}[s_{p_i} \mid c]$ and conditional variances $\sigma_i^2 = \mathrm{Var}(s_{p_i} \mid c)$. In a flat SAE with effective L_0 penalty weight $\lambda > 0$ per active code, absorbing subset $A_{abs} \subseteq \{1, \ldots, m\}$ into the child's decoder atom $\tilde{d}_c = u_c + \sum_{i \in A_{abs}} \alpha_i u_{p_i}$ has optimal coefficient $\alpha_i^* = \mu_i$ and expected conditional objective:*

$$\mathcal{J}(A_{abs}) = \mathbb{E}\big[\|x - \hat{x}\|_2^2 + \lambda \|z\|_0 \;\big|\; c\big] = \lambda(1 + m) + \sum_{i \in A_{abs}} \big(\sigma_i^2 - \lambda\big). \tag{2}$$

Consequently, parent p_i is absorbed if and only if $\sigma_i^2 < \lambda$. Ordering parents by conditional variance $\sigma_{(1)}^2 < \sigma_{(2)}^2 < \cdots < \sigma_{(m)}^2$, parents are absorbed in strictly increasing order of σ_i^2, producing selective partial absorption *whenever $\sigma_{(1)}^2 < \lambda < \sigma_{(2)}^2$.*

3 DR-SAE Architecture and Training Objective

Given centered residual-stream activation $\tilde{x} = x - b_{\mathrm{dec}} \in \mathbb{R}^d$, DR-SAE computes native sparse codes via a BatchTopK encoder [13, 12] and routes activations upward along a sparse, non-negative parent matrix $A \in \mathbb{R}_{\geq 0}^{n \times n}$:

$$z = \mathrm{BatchTopK}_k\big(W_{\mathrm{enc}}(x - b_{\mathrm{dec}}) + b_{\mathrm{enc}}\big), \qquad z_{\mathrm{eff}} = (I - A)^{-1} z = \sum_{h=0}^{H} A^h z, \qquad \hat{x} = R z_{\mathrm{eff}} + b_{\mathrm{dec}}. \tag{3}$$

Every 2,000 steps (after a 10% native-code warmup), the structure refresher ranks latents by exponential moving average firing frequency f_i and proposes at most $m = 8$ candidate parents per latent c from strictly higher-frequency latents ($f_p > f_c$) maximizing the product of conditional co-activation lift and positive cosine alignment with c's empirical conditional activation direction. All parameters ($W_{\mathrm{enc}}, b_{\mathrm{enc}}, b_{\mathrm{dec}}, R, A$) are trained jointly under a single differentiable objective combining 4-bank Matryoshka reconstruction [2], L_1 edge sparsity, whitened-space hierarchical orthogonality, and AuxK dead-latent recovery:

$$\mathcal{L} = \frac{1}{4} \sum_{b=1}^{4} \big\| x - R_{\leq b}\, z_{\mathrm{eff},\leq b} - b_{\mathrm{dec}} \big\|_2^2 + \lambda_A \|A\|_1 + \lambda_\perp \sum_{(p,c)\in E} \langle \tilde{r}_p, \tilde{r}_c \rangle^2 + \alpha \mathcal{L}_{\mathrm{AuxK}}, \tag{4}$$

where $\tilde{r}_i = \Sigma^{-1/2} r_i / \|\Sigma^{-1/2} r_i\|_2$ denotes whitened unit residual atoms, $\lambda_A = 3 \times 10^{-4}$, $\lambda_\perp = 0.05$, $\alpha = 1/32$, and $H = 4$. At evaluation, edges with $A_{p,c} < 0.05$ are zeroed to export a frozen DAG.

4 MPAbs Benchmark and Pre-Registered Evaluation Protocol

To evaluate multi-parent feature absorption with statistical power, we construct **MPAbs**:

- **Tier 1: Mixed-Topology Synthetic Testbed (Extension of SynthSAEBench [10]):** Sweeps multi-parent share $\rho \in \{0.0, 0.10, 0.25, 0.50\}$ and a $5 \times 5 \times 4$ Proposition 2 grid ($\sigma_1^2, \sigma_2^2 \in \{0.04, 0.12, 0.25, 0.45, 0.75\}$, $\lambda \in \{0.08, 0.18, 0.35, 0.60\}$), plus 36 structured hard negatives per configuration (transitive ancestors, correlation-only pairs, and incomplete parent subsets).
- **Tier 2: MPAbs Multi-Parent Token Suite on Pythia-160M:** Comprises 112 stratified cells crossing initial letter (14 letters), WordNet lexicographer noun category (12 categories), and capitalization case (`lowercase` vs. `TitleCase`), plus a dual-hypernym semantic multiple-inheritance set (`"bat"`, `"salmon"`, `"Paris"`, `"piano"`). Each cell contains ≥ 25 tokens split 60/20/20 into train/validation/test. Logistic regression probes achieve a mean validation accuracy of **98.7%** (exceeding the $\geq 90\%$ gate).
- **Tier 3: Pythia-70M Sparse Feature Circuits:** Evaluates 6-layer residual SAEs across 4 subject-verb agreement tasks from `feature-circuits` [11] (`simple_sv`, `rc_sv`, `pp_sv`, `npz_sv`), measuring the number of feature nodes required to recover 0.80 task faithfulness.

5 Experimental Results Across Research Questions (RQ1–RQ4)

5.1 RQ1: Multi-Parent Scaling and Proposition 2 Selective Absorption Order

Table 1 reports parent-set recovery as the multi-parent concept share ρ increases from 0.0 (pure tree) to 0.50. While H-SAE (tree prior, $m \leq 1$) achieves 100.0% recall at $\rho = 0.0$, its recall collapses by **33.80 percentage points** to 66.20% at $\rho = 0.50$ because it can represent at most one parent per child. In contrast, DR-SAE maintains **98.59%** recall at $\rho = 0.50$ (a drop of only **1.41 points**, satisfying H1's ≤ 5.0 point threshold) while rejecting 100.0% of hard negatives. Across the 80 off-diagonal cells of the Proposition 2 grid, the lower-variance parent $\sigma^2_{(1)} < \sigma^2_{(2)}$ is absorbed first in **92.5%** of cells (74/80), confirming Proposition 2 ($p = 1.2 \times 10^{-6}$).

Table 1: RQ1 & RQ3 Synthetic Mixed-Topology Scaling Across Multi-Parent Share ρ (Mean Over 3 Seeds).

Method	**Parent-Set Edge Recall (%) ↑**				**Parent-Set F1 ↑**		**Hard-Neg. Rej. (%) ↑**	**Drop (ρ: 0 → 0.5) ↓**
	$\rho = 0.0$	$\rho = 0.10$	$\rho = 0.25$	$\rho = 0.50$	$\rho = 0.25$	$\rho = 0.50$	($\rho = 0.25$)	(Recall Pts)
H-SAE (Tree Prior)	100.00	90.38	79.66	66.20	0.8228	0.6571	100.00	−33.80 pts
AG-SAE-lite (Post-hoc)	98.58	99.36	96.40	91.20	0.8440	0.7655	75.75	−7.38 pts
Post-hoc DR (Frozen Enc.)	100.00	99.10	97.80	94.50	0.8276	0.7770	69.87	−5.50 pts
DR-SAE (Ours)	**100.00**	**99.36**	**100.00**	**98.59**	**0.9420**	**0.9310**	**100.00**	**−1.41 pts**

5.2 RQ2 & RQ3: MPAbs Absorption, Matched Effective L_0, and Causal Readout

Table 2 presents the Phase 6 test-split results across all 6 methods plus DR-SAE at strictly matched effective $L_0 = 50.0$. Notably, H-SAE achieves 1.25% absorption on single-parent first letters but suffers **28.29%** mean MPAbs and 18.69% joint absorption because its single-parent routing leaves semantic and case parents unrouted. At matched native $L_0 = 50.0$, DR-SAE lowers mean MPAbs from 34.05% (Matryoshka) to **3.99%** (**88.3% relative reduction**, 95% bootstrap CI [84.1%, 91.8%]). Crucially, at strictly matched effective $L_0 = 50.0$

(native $k = 38.2$), DR-SAE achieves **6.71%** mean MPAbs (**80.3% relative reduction** vs. Matryoshka, 95% CI $[74.8\%, 85.1\%]$, paired Wilcoxon $W = 5996.0, p = 8.19 \times 10^{-19}$) while keeping $\Delta\text{FVU} = +0.0036 \leq 0.01$.

Table 2: Phase 6 Main Benchmark Results on Pythia-160M (MPAbs Test Split) and Pythia-70M Circuits (3 Seeds).

Method	Nat. L_0	Eff. L_0	FVU ↓	Loss Rec. ↑	1st-Let. Abs. ↓	MPAbs Mean ↓	MPAbs Joint ↓	MNN Repro. ↑	Causal ΔLogit ↑	Nodes@0.80 ↓
BatchTopK SAE	50.0	50.0	0.1362	0.9428	56.61%	57.26%	60.71%	—	1.256	133.8
Matryoshka BatchTopK	50.0	50.0	0.1318	0.9446	23.93%	34.05%	27.26%	—	1.796	102.7
H-SAE (Tree Prior)	50.0	56.4	0.1345	0.9435	1.25%	28.29%	18.69%	64.8%	2.369	94.9
AG-SAE-lite	50.0	61.2	0.1351	0.9433	5.95%	22.04%	12.92%	36.4%	2.402	86.6
Post-hoc DR	50.0	60.8	0.1358	0.9430	5.65%	25.97%	16.31%	54.2%	2.163	92.0
DR-SAE (Matched Eff. L_0)	38.2	**50.0**	0.1354	0.9431	**0.00%**	**6.71%**	**0.89%**	**76.2%**	**3.819**	**66.2**
DR-SAE (Ours, Nat. L_0)	**50.0**	63.8	0.1329	0.9442	**0.00%**	**3.99%**	**0.24%**	**78.4%**	**4.260**	**62.6**

To guard against the criticism that ancestral routing merely relabels inactive parents without causal content, our **Causal Readout Check** ablates each routed parent atom $z_{\text{eff},p} r_p$ on child tokens: DR-SAE produces a mean parent-probe logit drop of **4.260** (53.2× the random-atom ablation control of 0.080), compared to 1.256 in BatchTopK and 1.796 in Matryoshka. Furthermore, under Mutual Nearest Neighbour (MNN, cosine ≥ 0.70) alignment across seeds, DR-SAE recovers **78.4%** of edges across all 3 seeds, more than doubling AG-SAE-lite's **36.4%**.

5.3 RQ4: Downstream Sparse Feature Circuit Compactness and Ablations

When plugged into the 6-layer Sparse Feature Circuits pipeline on Pythia-70M across 4 subject-verb agreement tasks, DR-SAE requires an average of **62.6 feature nodes** to reach 0.80 faithfulness (and **66.2 nodes** at matched effective $L_0 = 50.0$), compared to **102.7 nodes** for Matryoshka (**39.0% reduction**) and **133.8 nodes** for BatchTopK (**53.2% reduction**), with mean faithfulness AUC rising from 0.843 (BatchTopK) and 0.880 (Matryoshka) to **0.927** (DR-SAE). Table 3 reports the 9 component ablations and Table 4 summarizes the Holm–Bonferroni hypothesis verification.

Table 3: DR-SAE Component Ablations on Pythia-160M MPAbs and Pythia-70M Circuits.

Ablation Variant	MPAbs Mean ↓	MPAbs Joint ↓	Synth. F1 ↑	FVU ↓	Nodes@0.80 ↓
Full DR-SAE ($m = 8, H = 4$, Freq. Order, Whitened $\perp$)	**3.99%**	**0.24%**	**0.942**	0.1329	**62.6**
Matched Effective L_0 ($k = 38.2 \implies$ Eff. $L_0 = 50.0$)	6.71%	0.89%	0.924	0.1354	66.2
Single-Parent Cap ($m = 1$, Tree Restriction)	20.79%	12.04%	0.654	0.1359	84.2
Matryoshka Bank Order Instead of Frequency Order	13.19%	6.64%	0.772	0.1349	75.4
No Orthogonality Loss ($\lambda_\perp = 0$)	11.39%	5.04%	0.858	0.1319	72.1
Raw-Space Orthogonality (Unwhitened)	6.59%	1.94%	0.914	0.1339	65.8
No Nested Matryoshka Reconstruction Loss	9.19%	3.84%	0.886	**0.1319**	69.5
Hop Depth $H = 1$ (Immediate Parents Only)	9.79%	4.34%	0.895	0.1349	70.3
Hop Depth $H = 8$ (Deep Unrolled Series)	**3.89%**	**0.24%**	**0.943**	0.1329	**62.4**

6 Limitations and Conclusion

Limitations: (1) All main LLM experiments use 3 random seeds on sub-200M parameter models (Pythia-160M and Pythia-70M); differences smaller than one cross-seed standard deviation (0.45% MPAbs) should be treated

Table 4: Pre-Registered Hypothesis Verification with Holm–Bonferroni Step-Down Correction ($\alpha = 0.05$).

ID	Pre-Registered Target	Empirical Outcome	Raw p-val	Holm p-val	Status
H2	$\geq 15\%$ MPAbs drop @ Eff. L_0	-80.3% (6.71% vs. 34.05%, ΔFVU $= +0.0036$)	8.19×10^{-19}	3.28×10^{-18}	**Confirmed**
H1	Tree drop ≥ 20 pts; Prop. 2 $\geq 80\%$	Tree -33.8 pts, DR -1.4 pts; Prop. 2 $= 92.5\%$	1.20×10^{-6}	3.60×10^{-6}	**Confirmed**
H4	$\geq 20\%$ fewer circuit nodes @ 0.80	-39.0% (62.6 vs. 102.7 nodes; -35.5% @ Eff. L_0)	8.50×10^{-6}	1.70×10^{-5}	**Confirmed**
H3	F1 ≥ 0.90; Repro. $\geq 60\%$ vs. $\leq 40\%$	Synth. F1 $= 0.942$; Repro. 78.4% vs. 36.4%	3.40×10^{-5}	3.40×10^{-5}	**Confirmed**

as inconclusive, and scaling to Gemma-2-2B is left as an immediate extension. (2) Circuit compactness is evaluated across 4 subject-verb agreement tasks; extending to Indirect Object Identification (IOI) and multi-step reasoning circuits will further test generalizability. (3) Under Lemma 1, if a parent and child are near-duplicates ($\beta \leq \frac{(1-\tau)f_c}{1-f_c}$) with noisy frequency estimates, frequency ordering can drop the edge, which aligns with AG-SAE's non-redundancy criterion.

Conclusion: TopoCircuit-SAE demonstrates that feature absorption in real language models is fundamentally a *multi-parent* phenomenon that single-parent metrics and tree-structured priors fail to resolve. By enforcing acyclicity through firing-frequency order and learning end-to-end ancestral routing over orthogonal residual atoms, DR-SAE eliminates over 80% of multi-parent absorption at strictly matched effective L_0 and yields 39% more compact causal feature circuits.

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